

1. Project Title

Explainable AI-Based Career Intelligence and Skill Gap Analysis Platform for Students and Early-Career Job Seekers (**AI-based Career Intelligence and Skill Gap Analysis Platform**)

2. Introduction

The rapid growth of digital recruitment platforms has made it easier for students and job seekers to discover employment opportunities. Platforms such as LinkedIn, Naukri, Indeed, and Internshala provide job search, profile creation, resume management, job recommendations, and other career-related services.

However, finding a suitable job is only one part of the career decision-making process. Students and fresh graduates often face questions such as: *Which career role is suitable for my current skills? What skills am I missing? Why am I not a strong match for a particular role? What should I learn next?*

The proposed project aims to develop an **AI-based Career Intelligence and Skill Gap Analysis Platform** that goes beyond basic job recommendation. The system will analyze a candidate's resume, education, skills, experience, projects, and career preferences. It will compare this profile with job and career-role requirements using Natural Language Processing (NLP) and Machine Learning techniques.

The system will provide explainable job recommendations, career-readiness scores, prioritized skill gaps, job-market skill trends, and personalized learning roadmaps.

3. Background

The recruitment process has increasingly become data-driven. Job descriptions contain large amounts of unstructured textual information, while resumes contain information about a candidate's skills, education, experience, projects, and achievements.

Natural Language Processing can be used to extract meaningful information from these documents. Machine Learning and similarity-based techniques can then be used to compare candidate profiles with job requirements.

Traditional keyword-based matching may identify exact skill matches, but it may not fully capture semantic relationships between terms. For example, a resume mentioning "**data visualization using Microsoft's BI platform**" may be relevant to a job requiring "**Power BI**", even if the exact phrase does not appear.

Therefore, NLP and semantic similarity techniques can potentially improve the understanding of resumes and job descriptions.

The proposed project combines these techniques with **career-readiness assessment and skill-gap analysis** to create a more comprehensive career decision-support system.

4. Problem Statement

Students and early-career job seekers often have difficulty determining which career roles and job opportunities best match their current skills and qualifications.

Existing recruitment platforms provide job-search and recommendation functionality, but users may still need to determine independently:

- Whether they are adequately prepared for a particular career role.
- Which required skills they currently possess.
- Which important skills they are missing.
- Which missing skills should be prioritized.
- Why they received a particular job-match score.
- What learning path could help them become better prepared for their target role.

Therefore, there is a need for an intelligent system that integrates **candidate profiling, NLP-based skill extraction, explainable job matching, career-readiness assessment, skill-gap prioritization, and personalized learning recommendations** within a single platform.

5. Existing System Comparison

The following is an **initial conceptual comparison** for your Week-1 document. During your literature/competitor research, you should verify each feature and cite the relevant official documentation.

Feature	LinkedIn	Naukri	Internshala	Proposed System
Job search	✓	✓	✓	✓
Job recommendations	✓	✓	✓	✓
Resume/profile	✓	✓	✓	✓
Resume analysis	Partial	Partial	Partial	✓
Skill extraction	Partial	Partial	Partial	✓
Career readiness score	Not identified as core feature	Not identified as core feature	Not identified as core feature	✓
Explainable job matching	Limited/varies	Limited/varies	Limited/varies	✓
Skill-gap analysis	Limited/varies	Limited/varies	Limited/varies	✓

Feature	LinkedIn	Naukri	Internshala	Proposed System
Skill-gap prioritization	Not identified as core feature	Not identified as core feature	Not identified as core feature	✓
Personalized learning roadmap	Partial	Partial	Partial	✓
Career-path comparison	Limited	Limited	Limited	✓
Job-market skill trends	Partial	Partial	Partial	✓
What-if career simulation	Not identified	Not identified	Not identified	✓

6. Proposed System

The proposed system is an AI-based career intelligence platform designed primarily for students, fresh graduates, and early-career job seekers.

The system will accept information such as:

- Resume
- Education
- Technical skills
- Experience
- Projects
- Certifications
- Career preferences

The system will use NLP to extract and normalize relevant skills.

Job descriptions and career-role requirements will also be analyzed.

The candidate profile will then be compared with job requirements using a combination of **skill matching, text similarity, and semantic similarity**.

The system will generate:

Career Recommendations

Example:

- Data Analyst — 89%
- Business Analyst — 82%

- BI Analyst — 79%
- Data Scientist — 65%

Skill-Gap Analysis

Example:

Target: Data Scientist

Existing:

- Python
- SQL
- Pandas

Missing/Priority:

- Statistics
- Machine Learning
- Scikit-learn

Personalized Learning Roadmap

Statistics



Machine Learning



Scikit-learn



Machine Learning Project

7. Scope

The scope of the project is to develop a **career decision-support system** for students, fresh graduates, and early-career job seekers.

The system will cover:

- User profile creation
- Resume upload
- Resume parsing
- NLP-based skill extraction
- Candidate skill profiling

- Job and career recommendation
- Career-readiness assessment
- Explainable job matching
- Skill-gap identification
- Skill-gap prioritization
- Learning recommendations
- Job-market skill analysis
- Interactive dashboard

Outside the Scope

The project will **not** attempt to reproduce the complete functionality of platforms such as LinkedIn or Naukri.

The following are outside the scope:

- Social networking
- Recruiter messaging
- Payroll
- Employee management
- Background verification
- Complete recruitment management
- Video interviews
- Guaranteed employment prediction

The project will provide **career decision support**, not guarantee that a candidate will receive a job.

8. Expected Outcome

At the end of the project, a functional web-based prototype is expected to be developed.

A user should be able to:

Upload Resume



AI analyzes profile



Skills extracted



Career profile generated



Suitable careers identified



Jobs recommended



Career readiness calculated



Skill gaps identified



Skills prioritized



Learning roadmap generated

The system is expected to provide:

Career Recommendations

Suitable career roles based on the candidate profile.

Job Recommendations

Relevant jobs based on available job data.

Career Readiness

A model-based readiness score with supporting factors.

Explainability

Reasons behind job/career matching.

Skill Gap

Missing and prioritized skills.

Learning Roadmap

Recommended sequence for improving relevant skills.

Job-Market Intelligence

Analysis of skills currently requested in the available job data.